



SHASU VATHANAN

SHASU VATHANAN - GEN AI - PRODUCT MANAGER

GEN AI ARCHITECT JOURNEY · PHASE-002

Python for Generative AI

The 14 concepts to master before touching an AI library

Phase	Phase-002 - Python Foundations
Folder	Phase-002-GenAI-Python-Toolkit
Format	14 Jupyter notebooks and one module example
Level	Beginner
Learning style	Read - run - edit - experiment
Core dependency	Python 3

Shasu Vathanan - GEN AI - Product Manager

SHASUVATHANAN.COM · [linkedin.com/in/shasuvathanan](https://www.linkedin.com/in/shasuvathanan)



1. Purpose

Generative AI tools may feel like magic, but the code behind them still depends on solid Python basics.

This phase teaches the Python foundations needed before working with Generative AI. The lessons use short explanations, relatable examples, and runnable Jupyter notebooks. Follow them in order, practise the code, and build the confidence to read, write, and debug the Python used in AI projects - without unnecessary jargon.

Source material

The notebook content in this phase is based on the open-source **Python Concepts** project by **Thirumurugan** (github.com/Thirumurugan240/Python_Concepts), studied and worked through as part of this journey.

2. Learning objectives

By the end of this phase you will be able to:

- understand and run basic Python programs;
- use variables, data types, operators, and collections;
- control program behaviour with conditions and loops;
- organise reusable logic with functions and modules;
- handle errors and work with files safely; and
- prepare a clean Python environment for future GenAI libraries and projects.

3. The learning path

Complete the notebooks in numerical order. Each one states why the concept matters once you start building with Generative AI.

No.	Concept	Why it matters for Gen AI
01	What Is Programming	How AI applications follow coded workflows
02	Python and Its Uses	Python drives AI models, APIs, data, and automation
03	Python Syntax	Clean syntax prevents failures in AI pipelines
04	Virtual Environments	Gen AI libraries often need specific package versions
05	Run Python in VS Code	A practical workspace for AI experiments
06	Operators	Calculations, filtering, and decision rules
07	Data Types	Prompts, scores, settings, and model responses use these
08	Lists, Sets, Tuples, Dictionaries	AI requests and responses are lists and dictionaries
09	Control Flow	Routes prompts, validates input, controls model behaviour
10	Nested If Statements	Multi-step validation and access logic
11	Loops	Processes batches of prompts, files, or model outputs
12	Functions	Keeps prompt and model workflows reusable and testable
13	Error Handling	Handles API, input, and runtime failures without crashing
14	File Handling	Saves prompts, responses, logs, and datasets



4. Environment setup

Step 1 - Install Python

Download Python from python.org/downloads. On Windows, select **Add Python to PATH** during installation, then verify:

```
BASH
python --version
pip --version
```

If **python** is not recognised on macOS or Linux, try **python3**.

Step 2 - Install Visual Studio Code and its extensions

Extension	When to install	Why it is needed
Python	After installing VS Code	Runs, formats, and debugs Python
Jupyter	After the Python extension	Opens and runs notebook cells

Step 3 - Create and activate a virtual environment

```
BASH
python -m venv venv
```

Platform	Command
Windows PowerShell	<code>.\venv\Scripts\Activate.ps1</code>
Windows Command Prompt	<code>venv\Scripts\activate.bat</code>
macOS or Linux	<code>source venv/bin/activate</code>

Leave the environment with **deactivate** when finished.

Step 4 - Install notebook support

```
BASH
python -m pip install --upgrade pip
python -m pip install ipykernel
```

In VS Code, open a notebook, click **Select Kernel**, and choose the **venv** Python environment.

5. Packages and tools

The lesson code mainly uses built-in Python features. No package is required for the core examples besides notebook support.

Package	Why it is needed	Install with
ipykernel	Runs notebook cells with the selected environment	<code>python -m pip install ipykernel</code>
requests	Optional - demonstrates installing a third-party library (Lesson 04)	<code>python -m pip install requests</code>

venv ships with Python and normally needs no separate installation.



6. How to work through it

1. Complete the environment setup above.
2. Open the phase folder in VS Code.
3. Start with **01_What_is_Programming.ipynb**.
4. Read each explanation, then run its code cells from top to bottom.
5. Change the example values and predict the output before running again.
6. Complete the notebooks in numerical order.
7. Run **python app.py** after Lesson 12 to explore a reusable module.
8. Build a mini project using functions, error handling, and file handling.

Learning tip

Do not only watch the code run. Break it, fix it, and remix it - that is where the real learning happens.

7. The module example

The phase includes a small example of splitting reusable code into another file:

```
PYTHON
from modules.addition import add

print(add(1, 2))
```

Run it from the phase root with **python app.py**; it prints **3**. This same pattern scales directly to Gen AI projects, where prompt templates, model calls, data processing, and utility functions are each kept in their own module.

8. Troubleshooting

Problem	Likely cause	Fix
python is not recognised	Python is missing from PATH	Reinstall with Add Python to PATH , or try python3
Notebook cells do not run	No kernel is selected	Install ipykernel and select the venv kernel
PowerShell blocks activation	Execution policy restriction	Run Set-ExecutionPolicy -Scope CurrentUser RemoteSigned , then activate again
ModuleNotFoundError	Package missing from the active environment	Activate venv , then python -m pip install <package>
modules cannot be imported	Run from the wrong folder	Open a terminal at the phase root and run python app.py

9. References



Resource	Link
Python documentation	docs.python.org/3
Python downloads	python.org/downloads
VS Code Python guide	code.visualstudio.com/docs/python/python-tutorial
Jupyter in VS Code	code.visualstudio.com/docs/datascience/jupyter-notebooks
Source repository	github.com/Thirumurugan240/Python_Concepts

Next

Phase-003-Robotic-Process-Automation-(RPA) puts this Python to work driving real software - desktop and browser automation with PyAutoGUI and Playwright.

Shasu Vathanan - GEN AI - Product Manager

SHASUVATHANAN.COM · linkedin.com/in/shasuvathanan